DTUMOS LLM Agents: Possible Use Cases

1. Taxi dispatch negotiations: The passenger and vehicle files already have trip details, schedules, and vehicle types. LLM-based agents can simulate how taxi companies balance fast pickups against profit, using real-time route calculations from module/osrm_routing.py and OSRM tiles in data/osrm/.
2. City policy testing: Teams of agents representing city planners, citizen groups, and budget analysts can test different rules (like subsidies or congestion fees) using module/dispatch.py and module/dispatch_cost.py logic, run simulations through module/simulator.py, and review results in simul_result/scenario_base/ before making recommendations.
3. Bike/scooter company competition: LLM-based agents for competing bike-share or scooter companies can use module/point_generator.py to suggest new station locations, negotiate operating zones to reduce wait times from passenger logs, while a regulator checks that disability access requirements are met.
4. Infrastructure planning: Teams of agents representing urban designers, accessibility advocates, developers, and budget reviewers can use passenger travel patterns and route data from OSRM outputs (data/osrm/seoul-latest.) to debate where to add wheelchair ramps, charging stations, or bus lanes, using simulation results as shared evidence.
5. Pricing strategy debates: Teams of agents can create test scenarios through module/simulator_helper.py, have marketing and operations teams debate surge pricing versus shared rides, then show the results through dashboard.py for citizen representatives to review.
6. Emergency response drills: An emergency commander agent handles road closures and reroutes trips through module/osrm_routing.py, while hospital and social service agents help mobility-limited passengers from the passenger dataset, all reporting to a mayor agent tracking emergency performance.
7. Environmental planning: Fleet managers, energy companies, and environmental watchdog agents review vehicle usage from vehicle_data.csv, test electric vehicle policies in simul_result outputs, and create compliance plans together.
8. Multi-city transit coordination: Logistics, public transit, and rideshare agents test connections between different transport types by adding multi-modal stops (using module/utils.py helpers), negotiate payment agreements, and check performance through repeated runs logged to visualization/dashboard/.